

Laporan Modul 6 — Web Server, Virtual Host, Reverse Proxy & TLS Internal

Workshop Administrasi Jaringan · PENS — Teknik Informatika dan Komputer

- Kelompok: X = 2 · srv01 = Web server + Reverse proxy
- OS: Debian 13.4 (trixie) · Tanggal: 2026-06-12
- Domain: app.lab2.local, portal.lab2.local (DNS Modul 5)

1. Ringkasan

srv01 dipasang NGINX dengan dua virtual host (portal statis & app reverse proxy), backend python sebagai systemd service, dan TLS internal memakai self-signed CA. Semua diuji dari cli01 via DNS.

2. Baseline & Virtual Host (Bagian A & 4)

- NGINX 1.26 aktif. Dua vhost di `sites-enabled`:
- `portal.lab2.local` → static `/srv/data/apps/portal` (ada ETag).
- `app.lab2.local` → reverse proxy.
- Default site dinonaktifkan. `nginx -t` sukses.
- Uji dari cli01: `curl http://portal.lab2.local` → "Portal Server"; `curl http://app.lab2.local` → "Backend App Running".

3. Reverse Proxy (5)

- Backend `python3 -m http.server 9000` (bind 127.0.0.1) dijalankan sebagai **systemd service demo-backend** (persisten).
- `app.lab2.local` memakai `proxy_pass http://127.0.0.1:9000` dengan header `Host` & `X-Real-IP`.
- Bukti: response app "Backend App Running" tanpa ETag (di-proxy).

Detail: `docs/reverse_proxy_design.md`. Evidence: `evidence/curl_http.txt`.

4. TLS Internal (6)

- Self-signed **Lab2 Internal CA** menandatangani cert `app.lab2.local` (dengan SAN). Dipasang di nginx `listen 443 ssl`.
- CA dipasang di trust store cli01 → `curl https://app.lab2.local` (tanpa -k) sukses. `openssl s_client : TLSv1.3, Verify return code: 0 (ok)`.

Detail: `docs/tls_internal_policy.md`. Evidence: `evidence/curl_https.txt`.

5. Logging, Hardening & Troubleshooting (7 & 8)

- Log per-vhost: `app_access.log` , `portal_access.log` (+ error logs) — terisi.
- `ss -tlnp` : nginx listen 80 & 443, backend 127.0.0.1:9000.
- Hardening: header `X-Content-Type-Options nosniff` & `X-Frame-Options DENY` ; default site dimatikan; backend tidak diekspos langsung.

Evidence: `evidence/nginx_test.txt` , `evidence/nginx_logs.txt` .

6. Analisis (jawaban)

Jawaban pertanyaan 1-4 tersebar di `docs/reverse_proxy_design.md` (web server vs reverse proxy; kenapa tidak ekspos port langsung) dan `docs/tls_internal_policy.md` (kenapa TLS wajib internal; risiko private key bocor).

7. Definition of Done (Checklist)

Item	Status
2 virtual host berjalan	OK (app, portal)
Reverse proxy aktif	OK ("Backend App Running")
HTTPS internal aktif	OK (TLSv1.3, verify 0)
Log terbaca	OK (per-vhost)
Repo lengkap	OK

8. Bukti Screenshot

[1] Virtual host via DNS dari cli01 (app reverse proxy + portal statis)

```
-- [1] Virtual host via DNS dari cli01 ==
-- app (reverse proxy) --
Backend App Running

-- portal (static vhost) --
<h1>Portal Server</h1><p>portal.lab2.local - Modul 6</p>
fishir@cli01:~$ _
```

[2] HTTPS/TLS internal dari cli01 (verified by internal CA)

```
== [2] HTTPS/TLS internal dari cli01 ==  
-- curl -k (abaikan verifikasi) --  
Backend App Running  
  
-- curl verified by internal CA --  
Backend App Running  
  
-- detail sertifikat --  
subject=C=ID, O=Lab2, CN=app.lab2.local  
issuer=C=ID, O=Lab2, CN=Lab2 Internal CA  
fishir@cli01:~$ _
```

[3] nginx srv01: config test, ports, vhost, logging

```
== [3] nginx srv01: config test, ports, vhost, log ==  
the configuration file /etc/nginx/nginx.conf syntax is ok  
configuration file /etc/nginx/nginx.conf test is successful  
ports: :443 :80 :9000  
vhosts: app.lab2.local portal.lab2.local  
app_access.log:  
10.20.2.1 - - [13/Jun/2026:02:17:17 +0700] "" 400 0 "-" "-"  
10.20.2.1 - - [13/Jun/2026:02:17:33 +0700] "GET / HTTP/1.1" 200 20 "-" "curl/8.14.1"  
fishir@srv01:~$ _
```

9. Penyimpangan dari Modul

- OS Debian 13 (trixie), bukan Debian 12.
- Backend dijalankan sebagai systemd service (bukan `python3 -m http.server` manual) agar persisten & reproducible.
- Cert TLS memakai SAN (wajib untuk verifikasi modern; modul tidak menyebut).
- User admin `fishir` menggantikan `student`.